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BIOMEDICAL AND HEALTH SCIENCES

Methods and applications for multi-study learning with genomics data

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Batch Effects in Genomic Data

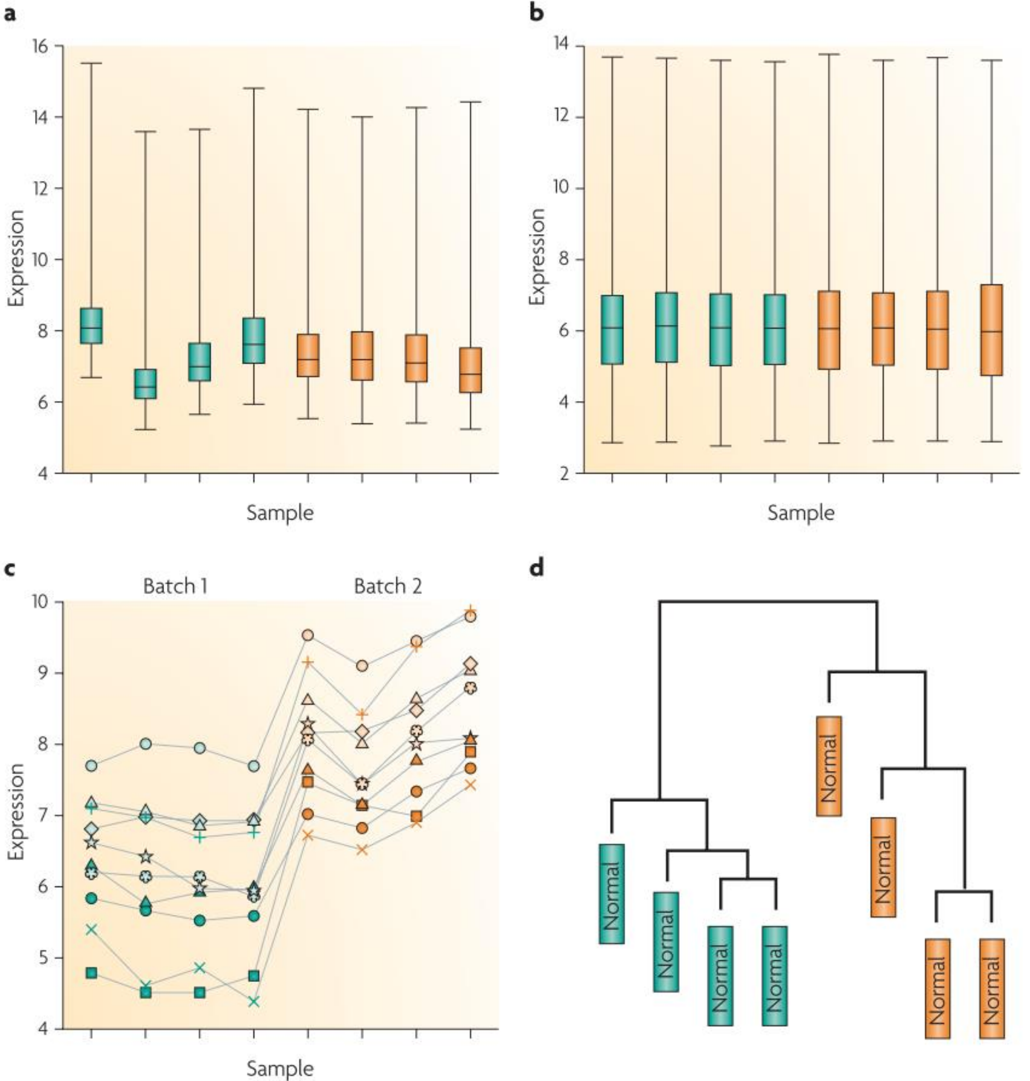
- ***Batch Effect:*** Non-biological variation due to differences in batches of data that confound the relationships between covariates of interest.

- Caused by differences:
  - Data profiling/generation platform
  - Lab protocol or experimenter
  - Time of day or hybridization
  - Atmospheric ozone level (Rhodes et al. 2004)

Batch Effect Example: bladder cancer

- Batch effect cannot be fully corrected using standard normalization methods
  - Overall adjustment does not remove all technical variation
  - Genes or cell types impacted differently by batch
  - Coverage differences
  - Library composition batch effect

Leek, Jeffrey T., et al. "Tackling the widespread and critical impact of batch effects in high-throughput data." Nature Reviews Genetics 2011



Combining Batches (ComBat) of Genomic Data

Consider the following model:

$$Y_{ijg} = \alpha_g + X\beta_g + \gamma_{ig} + \delta_{ig}\epsilon_{ijg}$$

where:

- α_g is the overall gene expression
- X is a design matrix
- β_g contains the regression coefficients
- The error terms $\epsilon_{ijg} \sim N(0, \sigma_g^2)$
- γ_{ig} and δ_{ig} are additive and multiplicative batch effects

Johnson, et al. "Adjusting batch effects in microarray expression data using empirical Bayes methods." Biostatistics, 2007

ComBat-Seq algorithm

Data — Count matrix

Step 1 Model
1. Fit Negative Binomial regression model
2. Obtain batch effect estimates

Step 2 Adjust
1. Compute "batch-free" distribution
2. Quantile adjustment

Yuqing Zhang

Zhang, et al. "ComBat-seq: batch effect adjustment for RNA-seq count data." NAR Genomics Bioinformatics., 2020

ComBat-Seq algorithm: Model

Negative Binomial regression

Gene-wise model: for a certain gene g, count in sample j from bat $y_{gij} \sim NB(\mu_{gij}, \phi_{gi})$

$$\log \mu_{gij} = \alpha_g + X_j\beta_g + \gamma_{gi} + \log N_j \qquad Var(y_{gij}) = \mu_{gij} + \phi_{gi}\mu_{gij}^2$$

Decompose scaled counts into 3 components

- α_g Average level for gene g (in “negative” samples)
- $X_j\beta_g$ Biological condition of sample j
- γ_{gi} Mean batch effect

ϕ_{gi} Variance batch effect

Yuqing Zhang

Zhang, et al. "ComBat-seq: batch effect adjustment for RNA-seq count data." NAR Genomics Bioinformatics., 2020

ComBat-Seq algorithm: Adjust

Adjust the data

Original count matrix

	S 1	S 2
G 1	14	12
G 2	0	0
G 3	112	11
…		

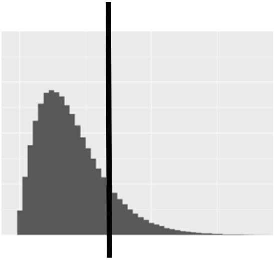
Adjusted count matrix

	S 1	S 2
G 1	9	8
G 2	0	0
G 3	60	47
…		

Yuqing Zhang

Empirical distribution of original counts:

$y_{gij} \sim NB(\hat{\mu}_{gij}, \hat{\phi}_{gi})$



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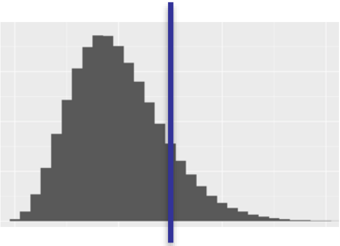


Batch-free distribution for adjusted counts:

$y^*_{gj} \sim NB(\mu^*_{gj}, \phi^*_g)$

Zhang, et al. "ComBat-seq: batch effect adjustment for RNA-seq count data." NAR Genomics Bioinformatics., 2020

BMC Part of Springer Nature

Genome Biology

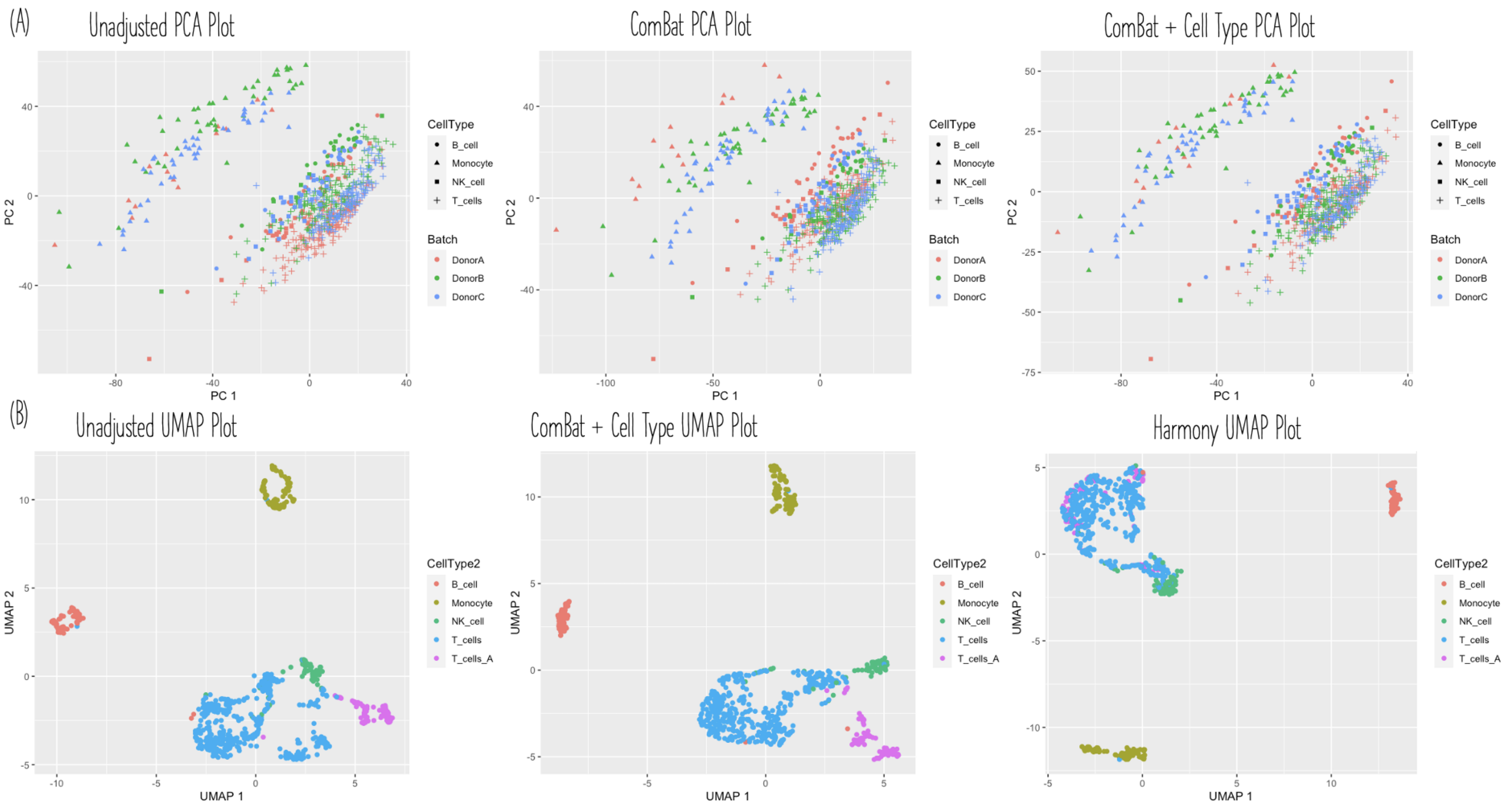
Research | Open Access | Published: 16 January 2020

A benchmark of batch-effect correction methods for single-cell RNA sequencing data

Hoa Thi Nhu Tran, Kok Siong Ang, Marion Chevrier, Xiaomeng Zhang, Nicole Yee Shin Lee, Michelle Goh & Jinmiao Chen

Genome Biology **21**, Article number: 12 (2020) | Cite this article

ComBat-Seq for single cell data

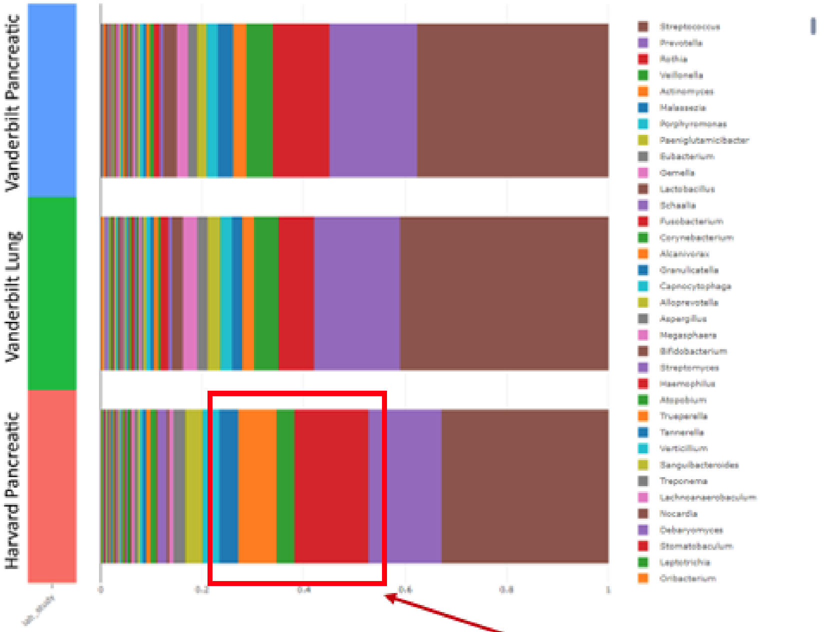


Raw Data Adjustment: Spike in Accuracy

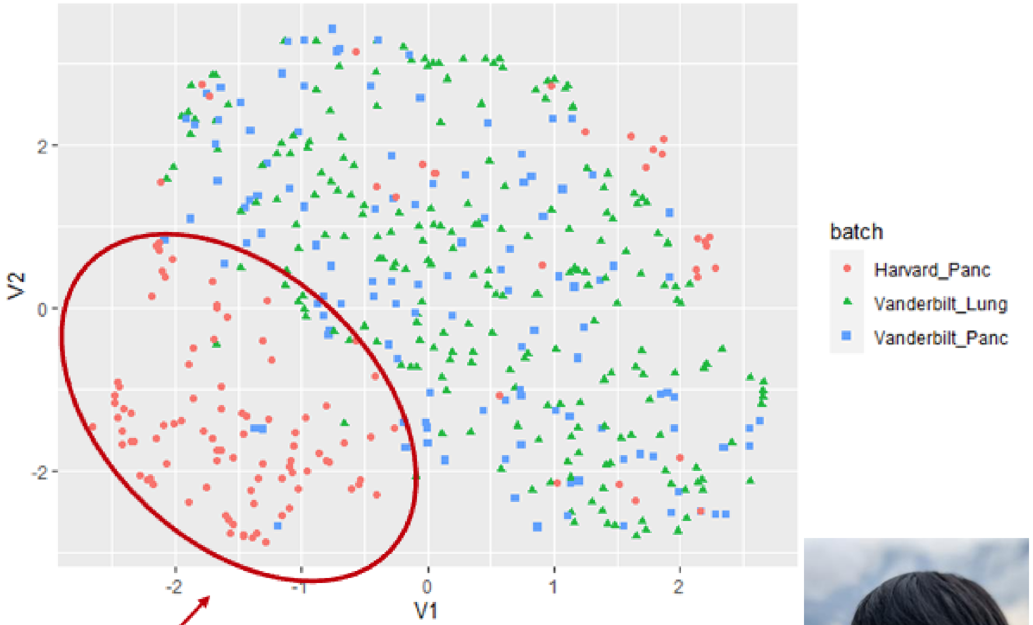
Method	Sensitivity	Specificity
Unadjusted data	0.319	0.869
ComBat-Seq	0.782	0.876
ComBat-Cell-Seq	0.841	0.849
ScMerge	0.718	0.782

GEE Models for Microbiome Analysis

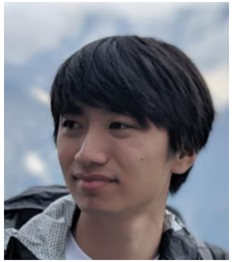
Relative Abundance Bar Plot



UMAP Plot



Potential Batch Effect



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RUTGERS GEE Models for Microbiome Analysis

- Over-dispersed, zero-inflated, high dimensional, compositional data
- For example, model for Batch effects:

$$Y_{ijg} \sim NB(\mu_{ijg}, \phi_{ig})$$

$$\log \mu_{ijg} = \alpha_g + X_j \beta_g + \gamma_{ig} + \log N_j$$

$$var(Y_{ijg}) = V_{ij}^{1/2} R_m V_{ij}^{1/2}$$

$$V_{ij} = diag_m(\mu_{ijm} + \phi_{im} \mu_{ijm}^2)$$

R_m describes compositional nature

- Methods and tools for longitudinal microbiome data

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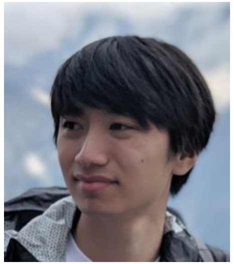
Brie Odom



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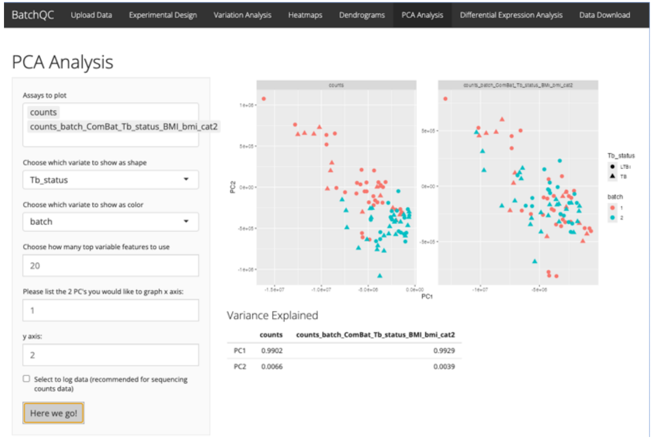
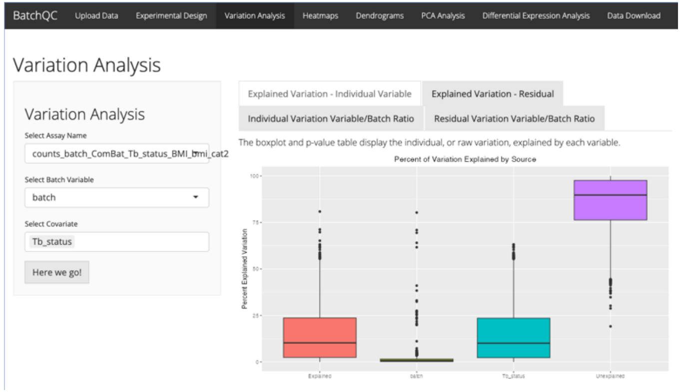
Lucas Schiffer

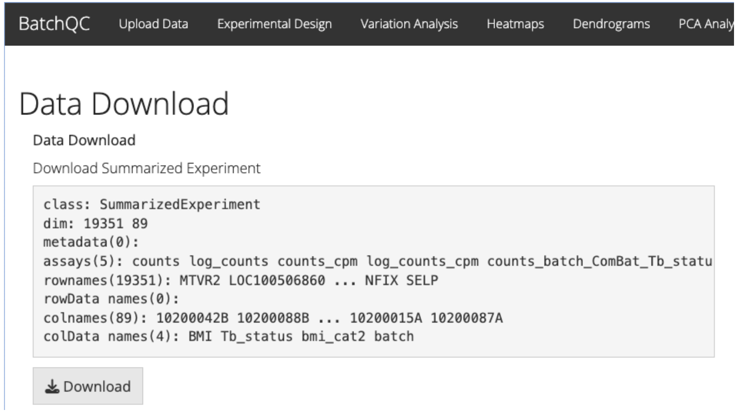
BatchQC: Evaluating Batch Effects

- Test for Negative Binomial Assumption
- Apply multiple batch correction methods (including ComBat-Seq)
- Evaluate need for batch
- Interactive Visualizations
- Compare original and batch corrected data side by side
- Export corrected data
- Shiny App or R functions



Jessica McClintock





Solaiappan Manimaran

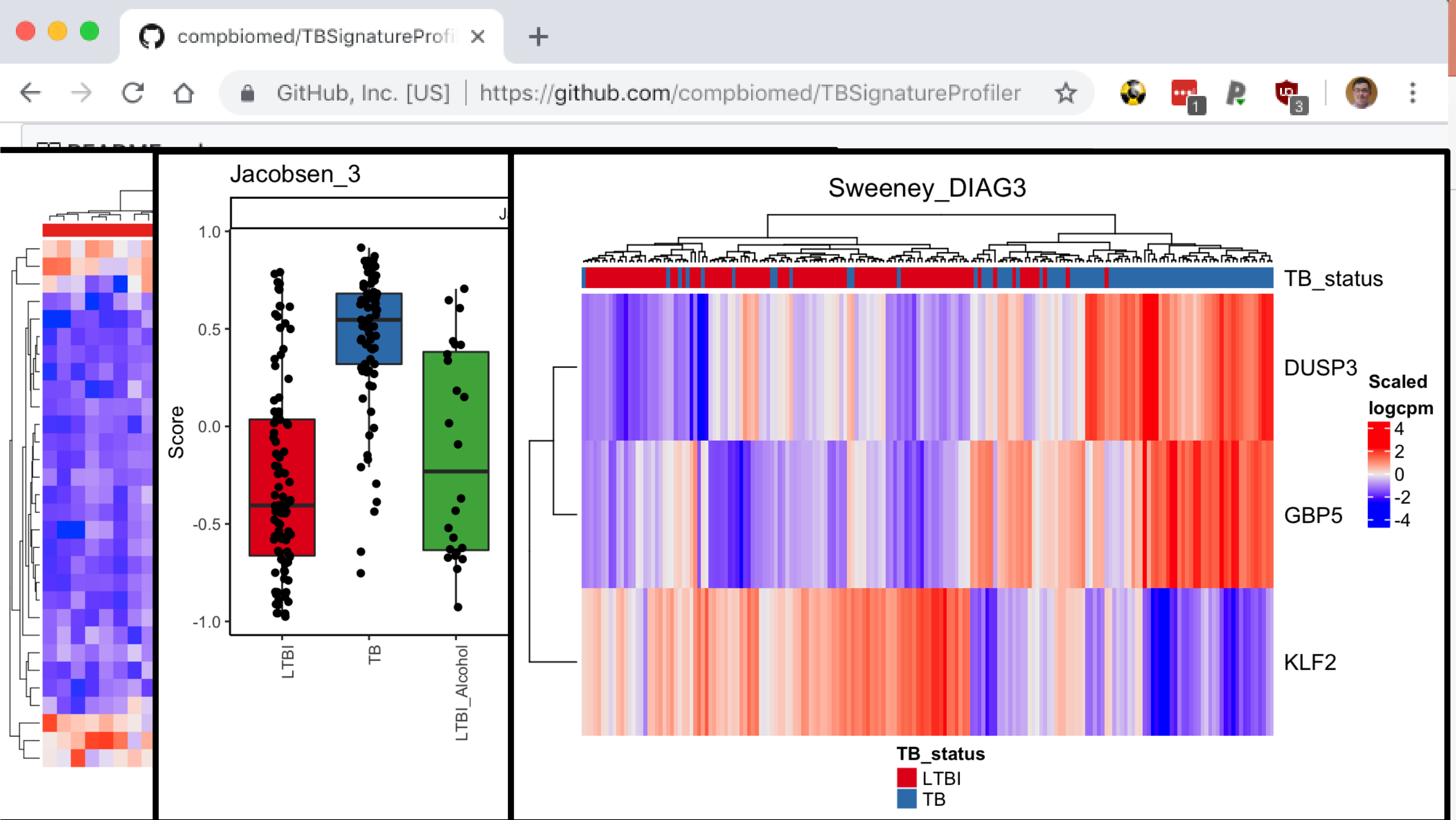
The TBSignatureProfiler

- R Package for profiling 70+ published signatures
  - Signatures of disease, response, and progression
- Scoring with original models (**new!**) and gene set enrichment methods:
  - GSVA, ssGSEA, PLAGE, singscore, zscore, and ASSIGN
- Visualization tools and functions:
  - Annotated heatmaps, boxplots, ROC, p-value and AUC tables
- **NEW!** TBSignatureProfiler graphical interface

Brie Odom

David Jenkins

Johnson et al, BMC Infectious Diseases, 2020; https://github.com/compbiomed/TBSignatureProfiler



compbiomed/TBSignatureProfi
GitHub, Inc. [US]
https://github.com/compbiomed/TBSignatureProfiler
Jacobsen_3
Score
1.0
0.5
0.0
-0.5
-1.0
LTBI
TB
LTBI_Alcohol
Sweeney_DIAG3
TB_status
DUSP3
GBP5
KLF2
Scaled logcpm
4
2
0
-2
-4
TB_status
LTBI
TB

The curatedTBData Project

- Curate expression data: currently 42+ TB experiments—more than 4,000 samples!
- Multiple platforms (array, sequencing, etc)
- Using SummarizedExperiment Objects in R
- Resource for biomarker validation/development!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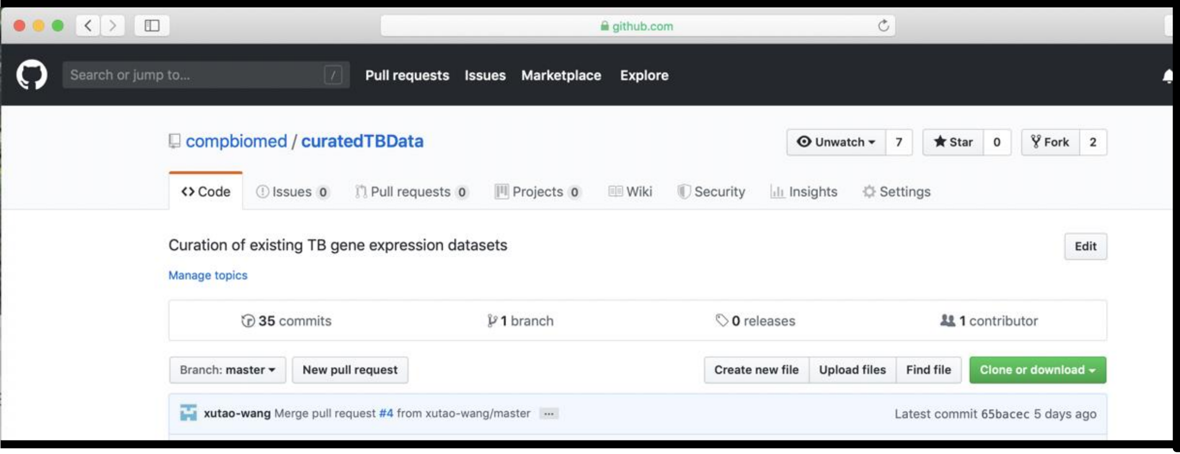
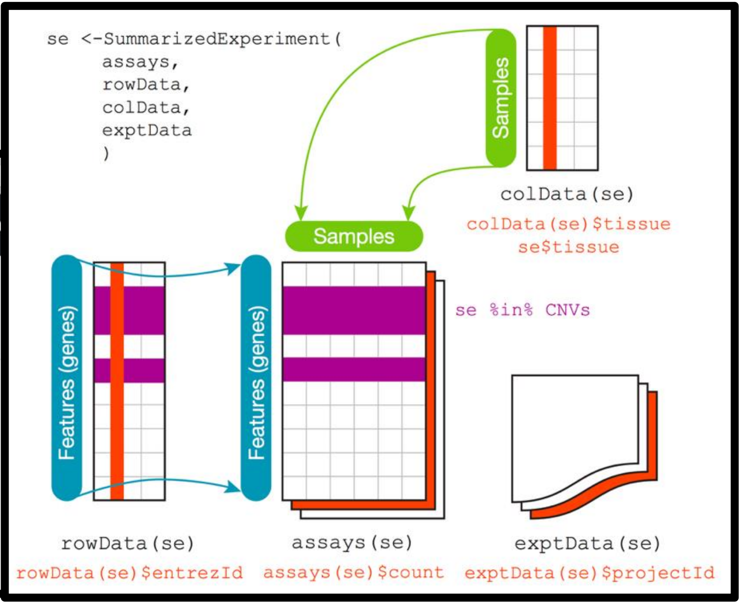
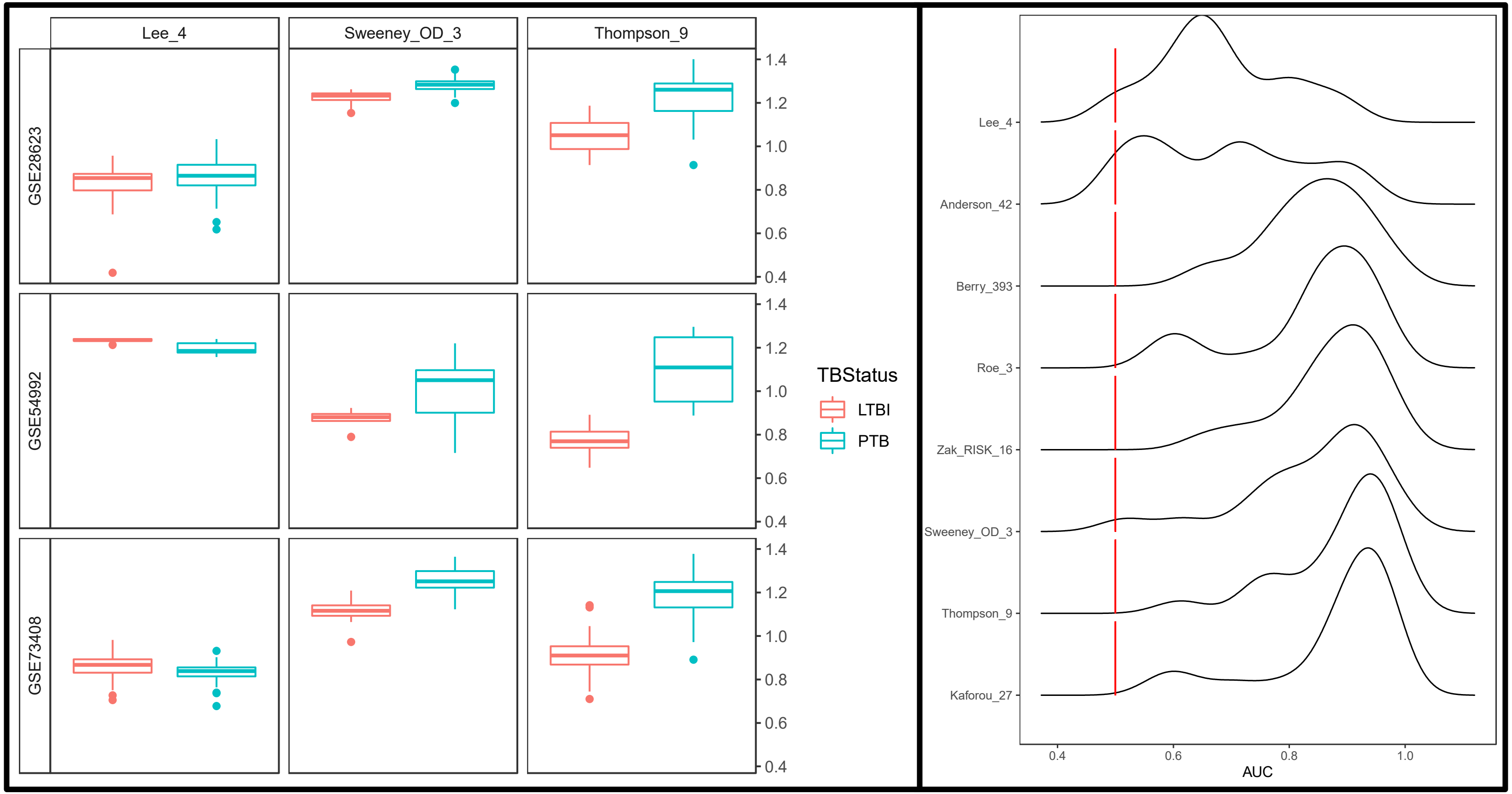




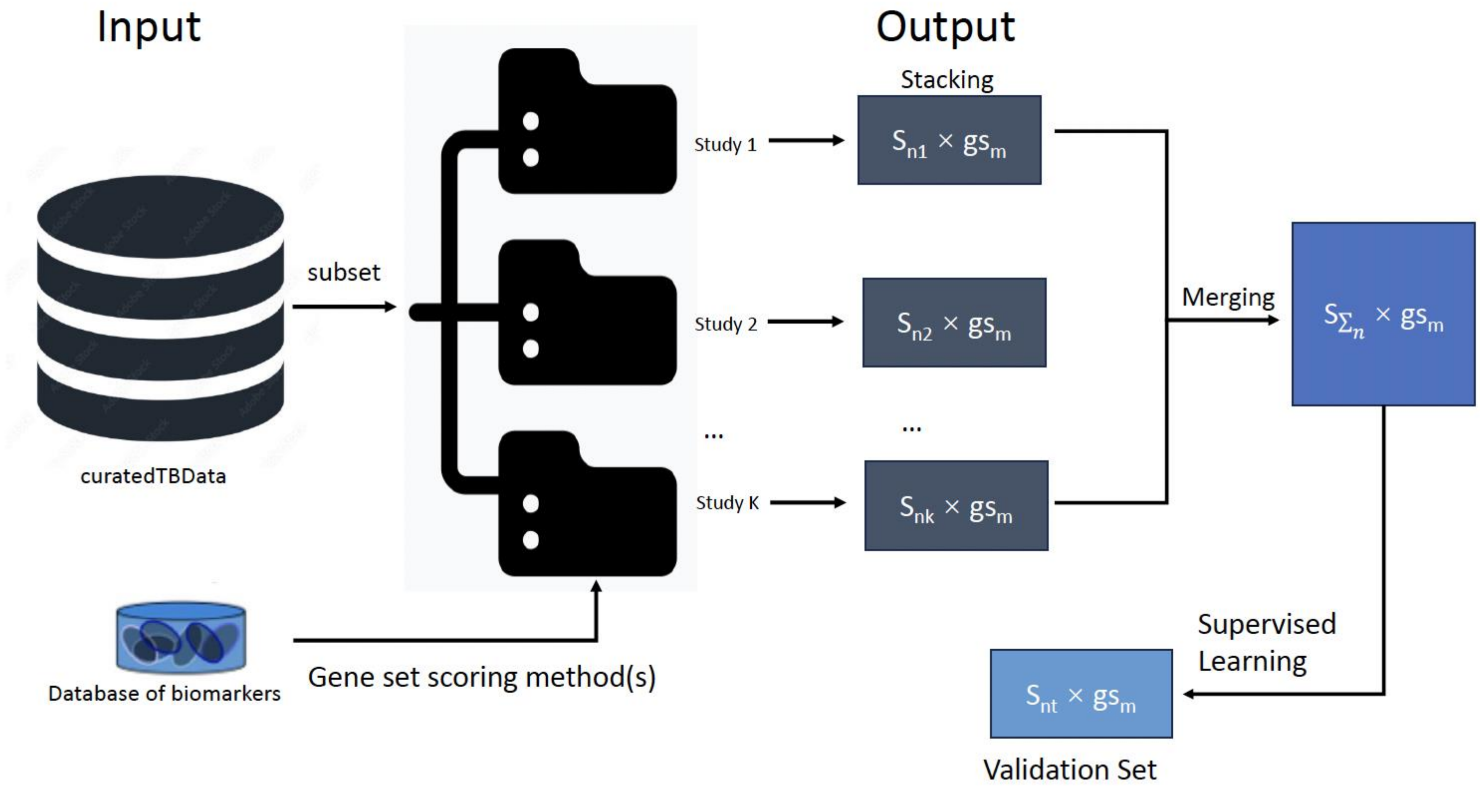
https://github.com/compbiomed/curatedTBData

curatedTBData and TBSignatureProfiler



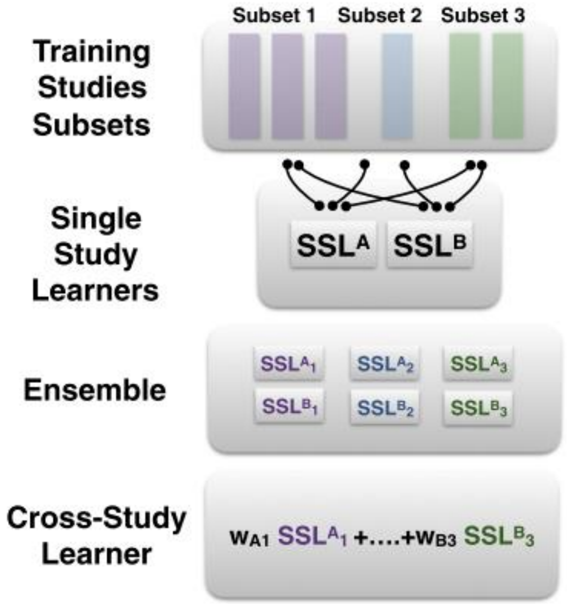
Multi-study Learning

Goal: Improve the predictive ability of existing biomarkers using cross study learning.



Solution when multiple studies available: ensemble learning

- Ensemble learning:
  - Established machine learning approach
  - Combined multiple models to improve prediction
- Cross-study learners (CSL)
  - Generalizable performance in independent test set
  - Experimental and theoretical insights (regression)



Cross-study learners from *Patil and Parmigiani, "Training replicable predictors in multiple studies", PNAS 2018*

Intuition

Merging VS Ensembling

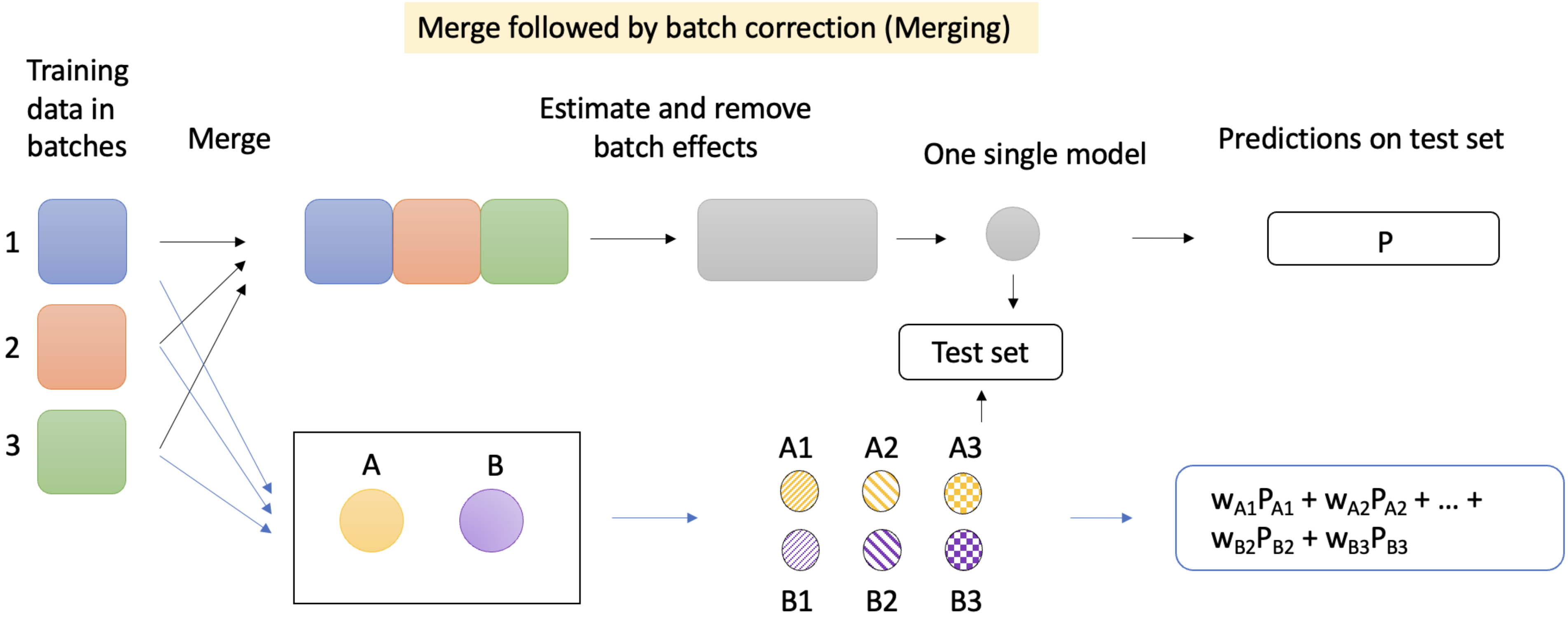
Merging

- Integrates data
- Remove undesired variation from as many of the genomic features as feasible, then use the "cleaned" data in training

Ensembling

- Integrates models
- Weights reward prediction functions that, while trained in one batch, continue to predict well in other batches
- likely to avoid using features affected by batch effect

Should we Merge or Ensemble?



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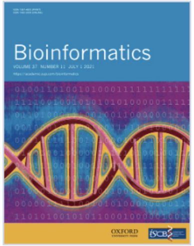

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Article Contents

JOURNAL ARTICLE

Robustifying genomic classifiers to batch effects via ensemble learning FREE

Yuqing Zhang, Prasad Patil, W. Evan Johnson, Giovanni Parmigiani ✉

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Abstract

Motivation

Yuqing Zhang

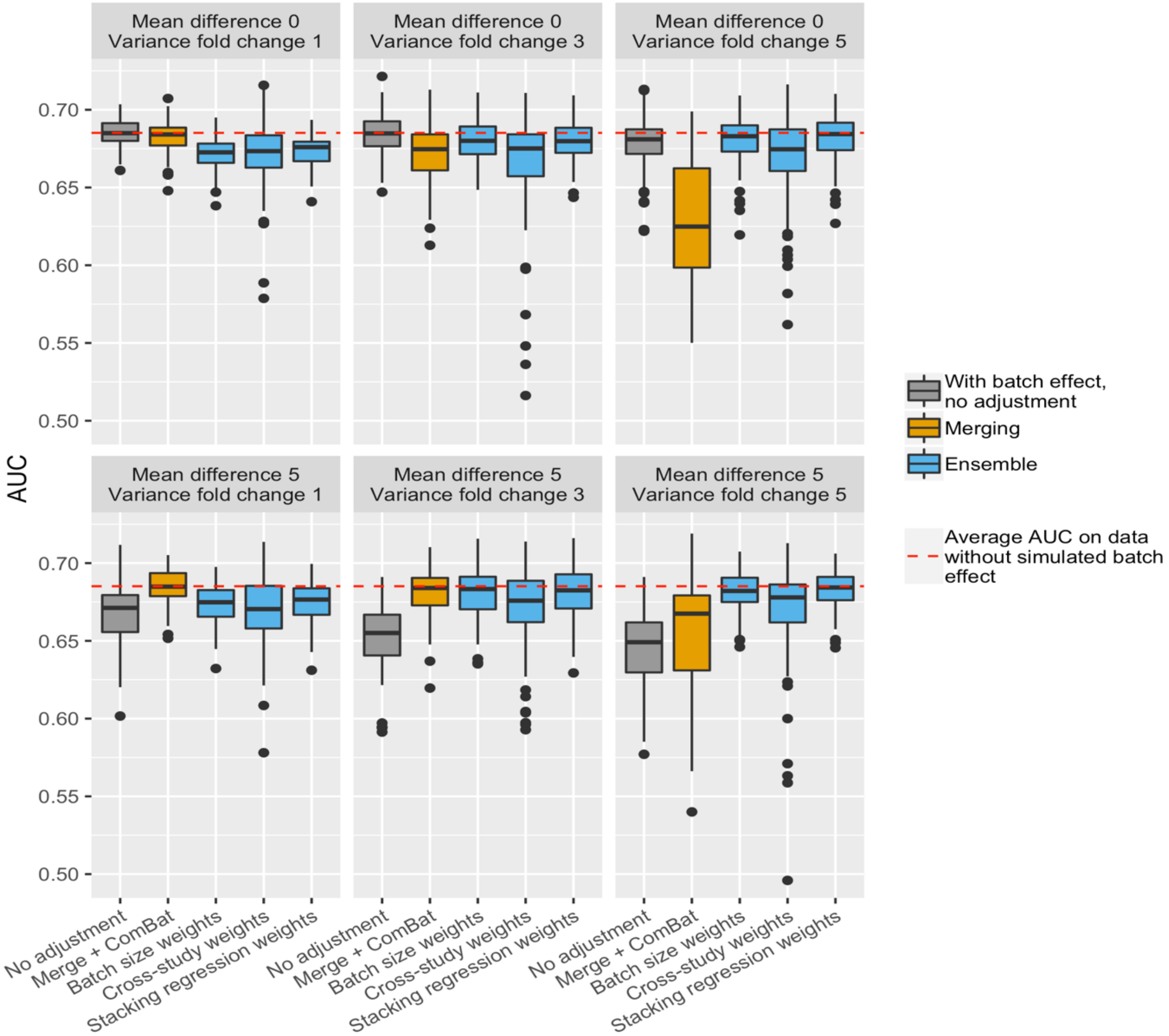
Prasad Patil

Giovanni Parmigiani

- Example results using random forest (RF) as learner

- *Transition point*: merging out-performs ensemble at low level of batch effect; ensemble is better at high batch differences

- Consistent with established cross-study results



Zhang, et al. "Robustifying genomic classifiers to batch effects via ensemble learning." Bioinformatics., 2021

Multi-Source Domain Adaptation

Traditional models:

Training and testing datasets are distributed identically. Violated by:

- covariate shift
- distribution gap

Multi-study learning

Common for multiple datasets with shared covariates and outcomes:

- Inefficient to rely on one data source for training

Multi-source domain adaptation:

- Merge all training studies
- Keep each training dataset separately

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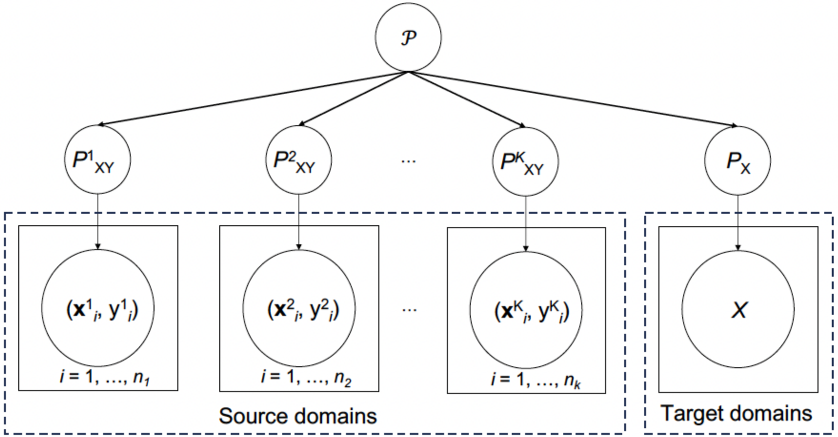
Multi-Source Domain Adaptation

- Denote $\mathcal{X}_S^j = \{(\mathbf{x}_i^j, y_i^j)\}_{i=1}^{n_j}$ as the *j-th* source/training domains, and $\mathcal{X}_T$ as target/testing domain with inaccessible label.
- Learn a robust and generalizable prediction function f from the K source domains s.t. $\min_f E_{(\mathbf{x},y)\in\mathcal{X}_T}[\ell(f(\mathbf{x}), y)]$.



(*Wang et al., IEEE Trans. Knowl. Data Eng., 2022*)

Integrating source-based classifiers:

$$y_i = \mathcal{F}(\mathbf{x_i}) = \sum_{j=1}^{K} w_j f_j(\mathbf{x_i}), \quad \sum_j w_j = 1$$

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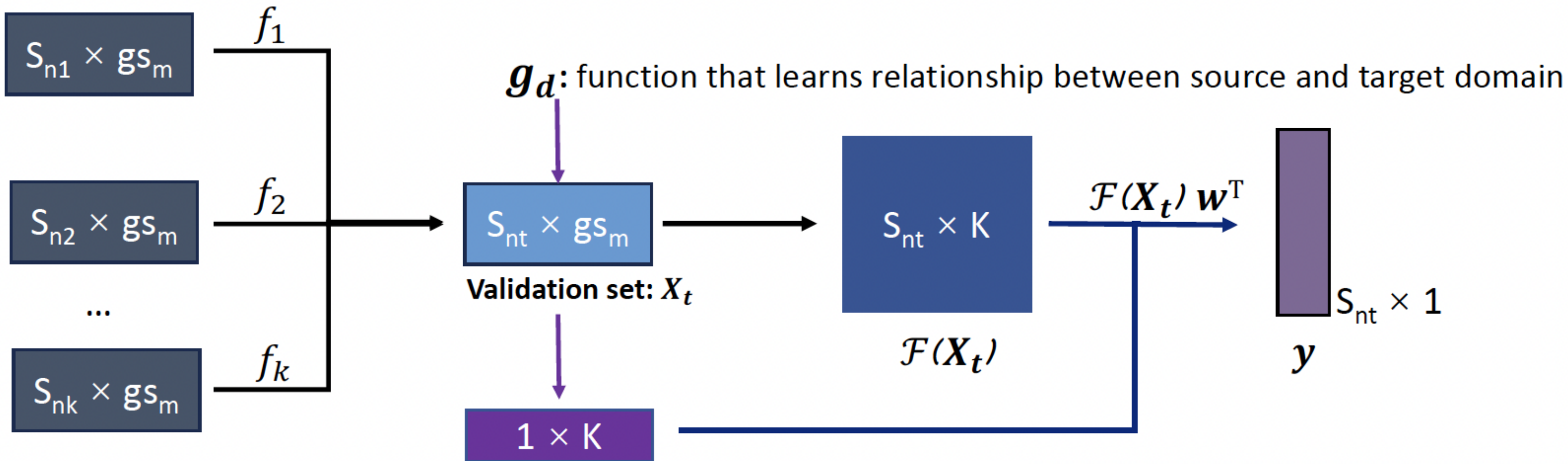
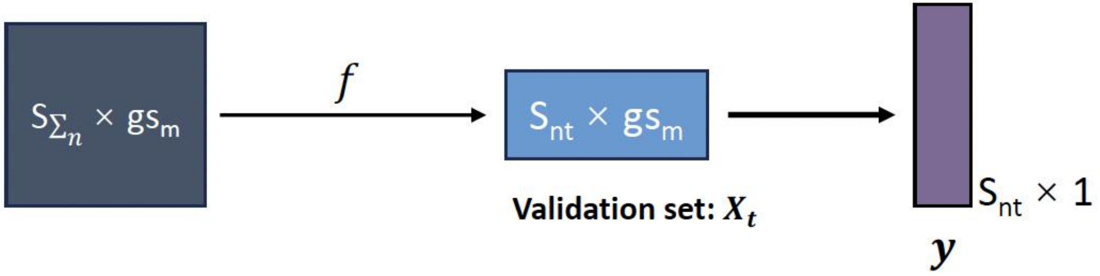
Multi-Source Domain Adaptation

Cross study learning:



Merging:



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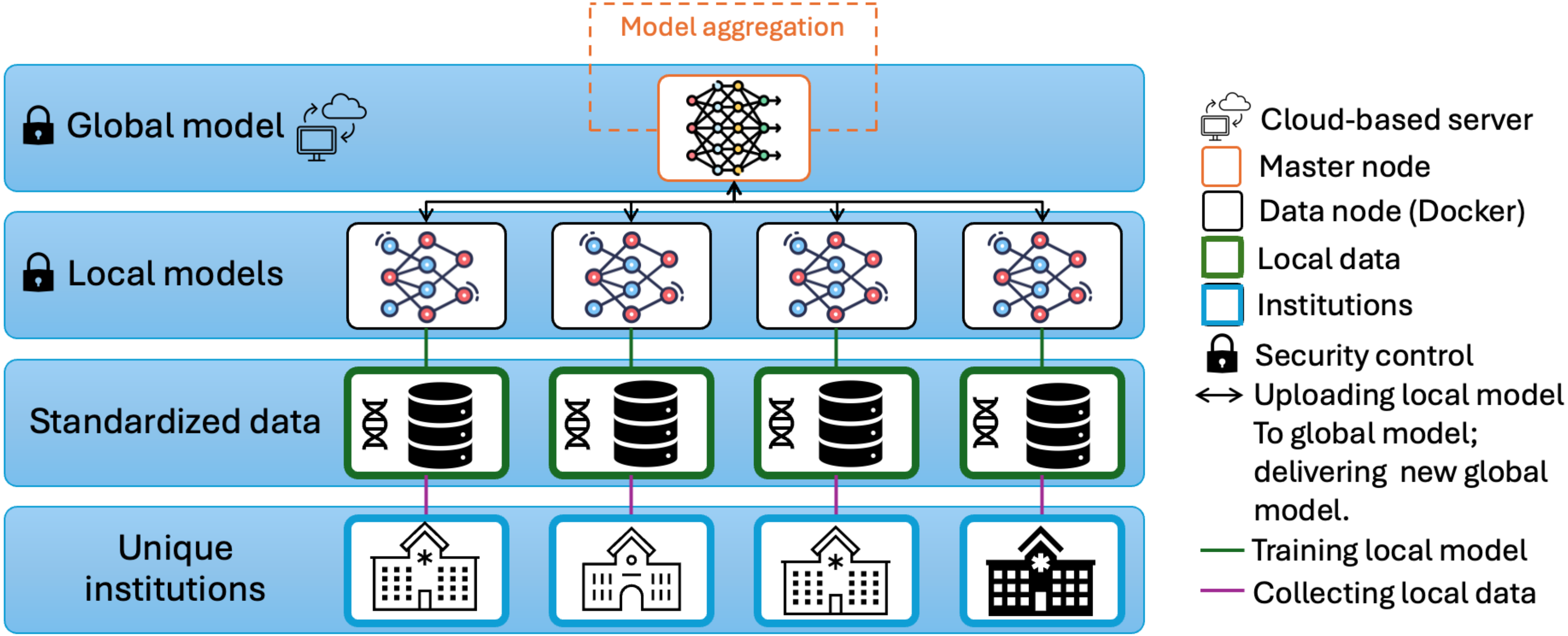
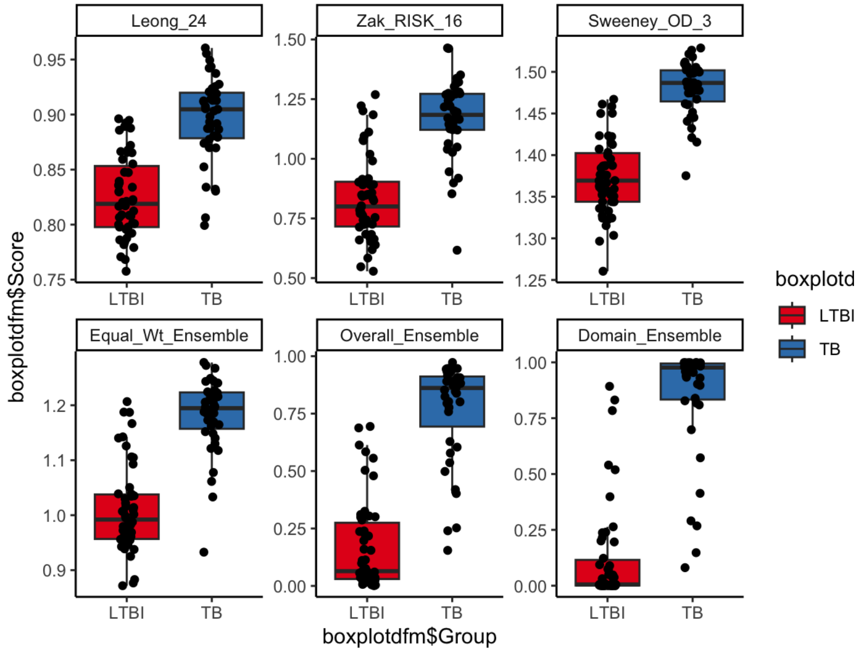
Adaptive Learning Improves Prediction

Comparison	Method	Accuracy	Sensitivity	Specificity
PTB vs. HCs	**Adaptive weight**	**0.84 (0.80-0.89)**	0.84 (0.76-0.92)	0.89 (0.83-0.93)
	Equal weight	0.84 (0.80-0.89)	0.84 (0.76-0.92)	0.89 (0.83-0.93)
	Sample-size weight	0.84 (0.80-0.90)	0.81 (0.72-0.90)	0.92 (0.89-0.96)
	Stacking	0.85 (0.80-0.91)	0.85 (0.78-0.92)	0.89 (0.84-0.95)
PTB vs. Others	**Adaptive weight**	**0.81 (0.77-0.84)**	0.77 (0.71-0.82)	0.85 (0.81-0.90)
	Equal weight	0.80 (0.76-0.84)	0.77 (0.72-0.82)	0.84 (0.79-0.89)
	Sample-size weight	0.75 (0.71-0.80)	0.56 (0.48-0.64)	0.92 (0.88-0.95)
	Stacking	0.76 (0.72-0.79)	0.56 (0.50-0.63)	0.93 (0.90-0.96)
PTB vs. OD	**Adaptive weight**	**0.69 (0.65-0.74)**	0.66 (0.57-0.76)	0.76 (0.71-0.81)
	Equal weight	0.67 (0.63-0.72)	0.65 (0.58-0.72)	0.75 (0.68-0.80)
	Sample-size weight	0.67 (0.62-0.72)	0.60 (0.51-0.69)	0.79 (0.73-0.83)
	Stacking	0.57 (0.48-0.67)	0.30 (0.13-0.53)	0.90 (0.83-0.96)

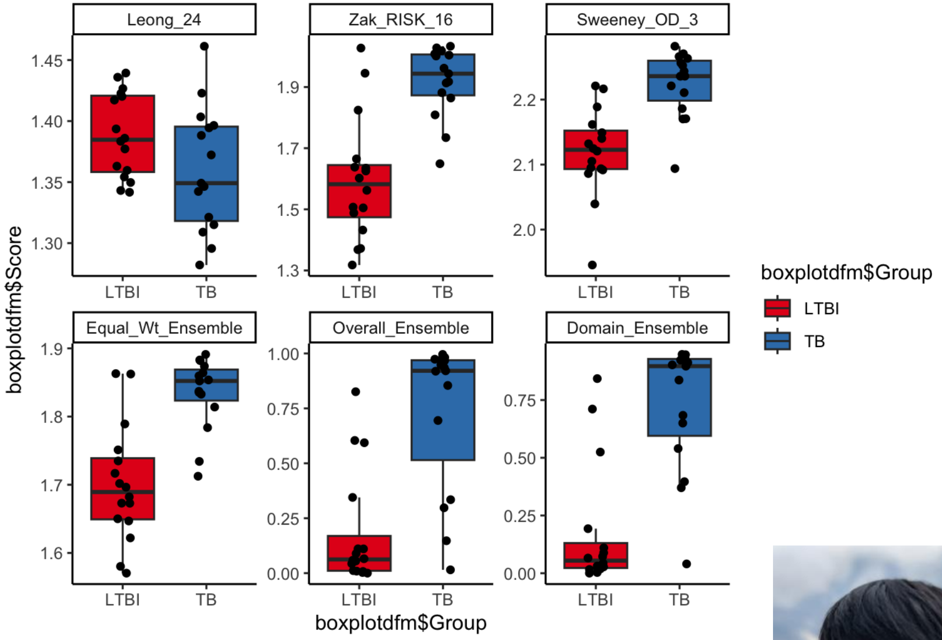
Federated Learning: Architecture



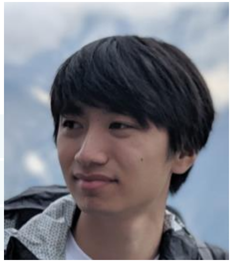
Indian Dataset



Ugandan Dataset



Country	Equal Weights	Ensemble Weights	Domain Weights
India	0.93	0.96	0.97
Africa	0.88	0.90	0.91



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